

- In each part, use the information in the table to find the dimension of the row space, column space, and nullspace of A , and of the nullspace of A^T .

	(a)	(b)	(c)	(d)	(e)	(f)	(g)
Size of A	3×3	3×3	3×3	5×9	9×5	4×4	6×2
$\text{Rank}(A)$	3	2	1	2	2	0	2

Question 04:

Step 1

1 of 3

(a)

Dimensions of the row and column spaces of A are equal to $\text{rank}(A) = 3$.

Dimension of the null space of A is $\text{nullity}(A)$. We know that

$$\text{rank}(A) + \text{nullity}(A) = n$$

where n is the number of columns of A . So

$$\text{nullity}(A) = n - \text{rank}(A) = 3 - 3 = 0$$

Also

$$\text{rank}(A^T) + \text{nullity}(A^T) = m$$

where m is the number of rows of A . Using $\text{rank}(A^T) = \text{rank}(A)$ we have

$$\text{rank}(A) + \text{nullity}(A^T) = m$$

Therefore dimension of the null space of A^T is

$$\text{nullity}(A^T) = m - \text{rank}(A) = 3 - 3 = 0$$

(b)

Dimensions of the row and column spaces of A are $\text{rank}(A) = 2$.

Dimension of the null space of A is

$$\text{nullity}(A) = n - \text{rank}(A) = 3 - 2 = 1$$

Dimension of the null space of A^T is

$$\text{nullity}(A^T) = m - \text{rank}(A) = 3 - 2 = 1$$

(c)

Dimensions of the row and column spaces of A are $\text{rank}(A) = 1$.

Dimension of the null space of A is

$$\text{nullity}(A) = n - \text{rank}(A) = 3 - 1 = 2$$

Dimension of the null space of A^T is

$$\text{nullity}(A^T) = m - \text{rank}(A) = 3 - 1 = 2$$

(d)

Dimensions of the row and column spaces of A are $\text{rank}(A) = 2$.

Dimension of the null space of A is

$$\text{nullity}(A) = n - \text{rank}(A) = 9 - 2 = 7$$

Dimension of the null space of A^T is

$$\text{nullity}(A^T) = m - \text{rank}(A) = 5 - 2 = 3$$

(e)

Dimensions of the row and column spaces of A are $\text{rank}(A) = 2$.

Dimension of the null space of A is

$$\text{nullity}(A) = n - \text{rank}(A) = 5 - 2 = 3$$

Dimension of the null space of A^T is

$$\text{nullity}(A^T) = m - \text{rank}(A) = 9 - 2 = 7$$

(f)

Dimensions of the row and column spaces of A are $\text{rank}(A) = 0$.

Dimension of the null space of A is

$$\text{nullity}(A) = n - \text{rank}(A) = 4 - 0 = 4$$

Dimension of the null space of A^T is

$$\text{nullity}(A^T) = m - \text{rank}(A) = 4 - 0 = 4$$

(g)

Dimensions of the row and column spaces of A are $\text{rank}(A) = 2$.

Dimension of the null space of A is

$$\text{nullity}(A) = n - \text{rank}(A) = 2 - 2 = 0$$

Dimension of the null space of A^T is

$$\text{nullity}(A^T) = m - \text{rank}(A) = 6 - 2 = 4$$

Result

3 of 3

- (a) Dimensions of the row and column spaces of A are 3. Dimension of the null space of A is 0. Dimension of the null space of A^T is 0.
- (b) Dimensions of the row and column spaces of A are 2. Dimension of the null space of A is 1. Dimension of the null space of A^T is 1.
- (c) Dimensions of the row and column spaces of A are 1. Dimension of the null space of A is 2. Dimension of the null space of A^T is 2.
- (d) Dimensions of the row and column spaces of A are 2. Dimension of the null space of A is 7. Dimension of the null space of A^T is 3.
- (e) Dimensions of the row and column spaces of A are 2. Dimension of the null space of A is 3. Dimension of the null space of A^T is 7.
- (f) Dimensions of the row and column spaces of A are 0. Dimension of the null space of A is 4. Dimension of the null space of A^T is 4.
- (g) Dimensions of the row and column spaces of A are 2. Dimension of the null space of A is 0. Dimension of the null space of A^T is 4.

5. In each part, find the largest possible value for the rank of A and the smallest possible value for the nullity of

(a) A is 4×4

(b) A is 3×5

(c) A is 5×3

Step 1

1 of 2

* Largest possible value of the *Rank* of any matrix of order $m \times n$ is $\min(m, n)$.

* Minimum possible value of the *Nullity* of any matrix of order $m \times n$ is No of columns of Matrix — Rank of a Matrix.

(a) A is 4×4

→ Maximum possible *Rank* of Matrix $A = \min(4, 4) = 4$

→ Smallest possible *Nullity* value of Matrix $A = 4 - 4 = 0$

(b) A is 3×5

→ Maximum possible *Rank* of Matrix $A = \min(3, 5) = 3$

→ Smallest possible *Nullity* value of Matrix $A = 5 - 3 = 2$

(c) A is 5×3

→ Maximum possible *Rank* of Matrix $A = \min(5, 3) = 3$

→ Smallest possible *Nullity* value of Matrix $A = 3 - 3 = 0$

6. If A is an $m \times n$ matrix, what are the largest possible value for its rank and the smallest possible value for its nullity?

Hint See Exercise 5.

Step 1

1 of 3

The goal of the exercise is to find the largest possible value of the rank of the $m \times n$ matrix

$$[A]_{m \times n}$$

and to find the smallest possible value for its nullity.

Step 2

2 of 3

Now since A is a $m \times n$ matrix therefore the row vectors of the matrix lie in the space $\mathbb{R}^n$ and the column vectors of the matrix lie in space $\mathbb{R}^m$.

We know that the rank of a matrix is the common dimension of its column space and row space. This leads us to conclude that the rank of the matrix A is less than or equal to the minimum of m and n , that is

$$\text{rank}(A) \leq \min\{m, n\}.$$

Thus we get the largest possible value of the rank of the matrix is

$$\min\{m, n\}.$$

Step 3

Now let's recall that the **Rank-Nullity theorem** states that if A is a $m \times n$ matrix then

$$\text{rank}(A) + \text{nullity}(A) = n.$$

Using this theorem we get the nullity of the matrix A is equal to

$$\begin{aligned} \text{nullity}(A) &= n - \text{rank}(A) \\ &\geq n - \min\{m, n\}. \end{aligned}$$

Thus we get the smallest possible value of the matrix A is

$$n - \min\{m, n\}.$$

- In each part, use the information in the table to determine whether the linear system $A\mathbf{x} = \mathbf{b}$ is consistent. If so, state the number of parameters in its general solution.

	(a)	(b)	(c)	(d)	(e)	(f)	(g)
Size of A	3×3	3×3	3×3	5×9	5×9	4×4	6×2
$\text{Rank}(A)$	3	2	1	2	2	0	2
$\text{Rank } [A \mid \mathbf{b}]$	3	3	1	2	3	0	2

Step 1

1 of 3

(a)

Because $\text{rank}(A) = \text{rank}[A|\mathbf{b}] = 3$ the system is **consistent**. The number of parameters in general solution is equal to $\text{nullity}(A)$. We know that

$$\text{rank}(A) + \text{nullity}(A) = n$$

where n is the number of columns of A . Therefore the number of parameters in general solution is

$$\text{nullity}(A) = n - \text{rank}(A) = 3 - 3 = 0$$

(b)

Because $\text{rank}(A) \neq \text{rank}[A|\mathbf{b}]$ the system is **not consistent**.

(c)

Because $\text{rank}(A) = \text{rank}[A|\mathbf{b}] = 1$ the system is **consistent**. The number of parameters in general solution is

$$\text{nullity}(A) = n - \text{rank}(A) = 3 - 1 = 2$$

Step 2

2 of 3

(d)

Because $\text{rank}(A) = \text{rank}[A|\mathbf{b}] = 2$ the system is **consistent**. The number of parameters in general solution is

$$\text{nullity}(A) = n - \text{rank}(A) = 9 - 2 = 7$$

(e)

Because $\text{rank}(A) \neq \text{rank}[A|\mathbf{b}]$ the system is **not consistent**.

(f)

Because $\text{rank}(A) = \text{rank}[A|\mathbf{b}] = 0$ the system is **consistent**. The number of parameters in general solution is

$$\text{nullity}(A) = n - \text{rank}(A) = 4 - 0 = 4$$

(g)

Because $\text{rank}(A) = \text{rank}[A|\mathbf{b}] = 2$ the system is **consistent**. The number of parameters in general solution is

$$\text{nullity}(A) = n - \text{rank}(A) = 2 - 2 = 0$$

Result

- (a) The system is consistent. The number of parameters in general solution is 0.
- (b) The system is not consistent
- (c) The system is consistent. The number of parameters in general solution is 2.
- (d) The system is consistent. The number of parameters in general solution is 7.
- (e) The system is not consistent
- (f) The system is consistent. The number of parameters in general solution is 4.
- (g) The system is consistent. The number of parameters in general solution is 0.

For each of the matrices in Exercise 7, find the nullity of A , and determine the number of parameters in the general solution of the homogeneous linear system $A\mathbf{x} = \mathbf{0}$.



Step 1

(a)

The number of parameters in general solution is equal to $\text{nullity}(A)$. We know that

$$\text{rank}(A) + \text{nullity}(A) = n$$

where n is the number of columns of A . Therefore the number of parameters in general solution is

$$\text{nullity}(A) = n - \text{rank}(A) = 3 - 3 = 0$$

(b) The number of parameters in general solution is: $\text{nullity}(A) = n - \text{rank}(A) = 3 - 2 = 1$

(c)

The number of parameters in general solution is

$$\text{nullity}(A) = n - \text{rank}(A) = 3 - 1 = 2$$

(d)

The number of parameters in general solution is

$$\text{nullity}(A) = n - \text{rank}(A) = 9 - 2 = 7$$

(e) The number of parameters in general solution is: $\text{nullity}(A) = n - \text{rank}(A) = 9 - 2 = 7$

(f)

The number of parameters in general solution is

$$\text{nullity}(A) = n - \text{rank}(A) = 4 - 0 = 4$$

(g)

The number of parameters in general solution is

$$\text{nullity}(A) = n - \text{rank}(A) = 2 - 2 = 0$$

Result

(a) 0

(b) 1

(c) 2

(d) 7

(e) 7

(f) 4

(g) 0

15. Prove: If $k \neq 0$, then A and kA have the same rank.

Proof. Let A be any matrix and let $k \neq 0$ be a scalar. We wish to show that $\text{rank}(kA) = \text{rank}(A)$.

Since $k \neq 0$, we may apply the following elementary row operation to the matrix kA :

$$R_i \leftarrow \frac{1}{k} R_i \quad \text{for each row } i.$$

Performing this operation on every row scales each entry of kA by $\frac{1}{k}$, which transforms kA into A :

$$\frac{1}{k}(kA) = A.$$

Since the transformation from kA to A is achieved entirely through elementary row operations, it follows by **Theorem 5.5.4** (elementary row operations do not change the row space of a matrix) that:

$$\text{row space}(kA) = \text{row space}(A).$$

Since rank is defined as the dimension of the row space, we conclude:

$$\text{rank}(kA) = \text{rank}(A).$$